—— AZ-104 // STORAGE

Configure Azure Files

SMB and NFS file shares in the cloud — share tiers, authentication methods, Azure File Sync, and how to mount shares on Windows, Linux, and macOS without touching a NAS.

Why Azure Files exists

Azure Files gives you a fully managed SMB or NFS file share in the cloud — no file server to patch, no NAS hardware to replace. Applications that need a shared drive letter, lift-and-shift workloads expecting a UNC path, and hybrid setups that need on-premises servers and cloud VMs reading the same files all land here instead of Blob Storage.

Share types — two protocols, two storage layers

SMB — STANDARD

Transaction Optimized

HDD-backed. Best for general-purpose file shares where throughput matters more than latency.

→ Supports SMB 2.1 / 3.x and REST

SMB — STANDARD

Hot

HDD-backed, optimized for general-purpose team file shares that are accessed regularly.

→ Supports SMB 2.1 / 3.x and REST

SMB — STANDARD

Cool

HDD-backed, optimized for online archive storage — lower storage cost, higher access cost.

→ Supports SMB 2.1 / 3.x and REST

SMB / NFS — PREMIUM

Premium (FileStorage)

SSD-backed, single-digit millisecond latency. Required for NFS shares and high-performance workloads. Billed by provisioned size, not usage.

→ Requires a FileStorage account (not GPv2)

Key rule: NFS shares require Premium tier and a FileStorage storage account — NFS is not available on Standard / GPv2 accounts.

Authentication methods

OPTION 1

Storage Account Key

Full access to all shares — equivalent to root. Simplest but least secure; avoid in production where possible.

OPTION 2 — RECOMMENDED

On-Prem AD DS (Kerberos)

Join the storage account to your on-premises Active Directory. Users authenticate with existing domain credentials over SMB.

OPTION 3

Entra ID Kerberos

Cloud-only AD authentication for Azure AD-joined or hybrid-joined machines. No on-premises AD required.

OPTION 4

Entra Domain Services

Managed AD DS in the cloud — authentication for machines joined to Entra DS when no on-premises AD exists.

Eight terms you need cold

File Share

A named SMB or NFS share within a storage account, accessible via a UNC path (`\\storageacct.file.core.windows.net\sharena` or a Linux mount point.

Tip: Quota sets the maximum size of the share — you can't write past it even if the underlying account has room.

Azure File Sync

A service that turns Azure Files into a cloud cache for an on-premises Windows Server — the server holds a local cache of hot files; cold files are replaced with cloud-tiering stubs.

Tip: File Sync requires a Windows Server 2012 R2 or newer with the File Sync agent installed and a registered Sync Group.

Sync Group

Defines the replication topology: one Azure file share (the cloud endpoint) plus one or more server endpoints (local paths on Windows Servers).

Tip: A server endpoint path must be on an NTFS volume — ReFS and FAT are not supported.

Cloud Tiering

An optional File Sync feature that replaces infrequently accessed files on the server endpoint with lightweight pointer stubs, freeing local disk space while keeping files accessible.

Tip: When a user opens a stub, it's transparently recalled from Azure — looks seamless but requires network access.

Share Snapshot

A read-only point-in-time copy of an Azure file share. Snapshots are incremental and stored within the same share.

Tip: Snapshots don't replace backups — use Azure Backup for file shares for policy-driven, geo-replicated protection.

SMB Encryption

Azure Files enforces SMB 3.0 encryption in transit by default — SMB 2.1 (unencrypted) is blocked unless explicitly allowed, which is not recommended.

Private Endpoint

A private IP inside a VNet that routes traffic to the file share without traversing the public internet — required for NFS shares, recommended for SMB in production.

Share-Level Permissions

RBAC roles (`Storage File Data SMB Share Reader/Contributor/Elevated Contributor`) that control access to the share as a whole — separate from NTFS directory/file-level permissions inside the share.

Tip: Both layers are evaluated — a user needs the share-level role AND appropriate NTFS ACLs to access a file.

SCENARIO	RIGHT CHOICE
Windows apps needing a shared drive letter	Azure Files SMB — Standard or Premium
Linux workloads needing a shared POSIX-compliant FS	Azure Files NFS — Premium (FileStorage account)
Extend on-prem file server to the cloud	Azure File Sync with cloud tiering
Object storage (unstructured data, blobs)	Azure Blob Storage — not Azure Files
High-performance databases, VM disks	Azure Managed Disks or Ultra Disk — not Azure Files

EXAM GOTCHA

NFS shares require **Premium tier** and a **FileStorage account type** — you cannot create an NFS share on a General Purpose v2 (GPv2) account. SMB shares work on both account types.

Setting up an Azure file share

Walk through this when deploying a new file share or migrating an on-premises file server.

START

You need a shared file system accessible by multiple machines

DECIDE

What protocol does the workload need?

SMB (WINDOWS / LINUX)

Standard GPv2 or Premium FileStorage account

NFS (LINUX / POSIX)

Must use Premium FileStorage account only

DECIDE

What performance tier is required?

GENERAL USE

Standard — Transaction Optimized, Hot, or Cool

LOW LATENCY / HIGH IOPS

Premium — SSD-backed FileStorage account

DECIDE

How will users authenticate to the share?

ON-PREM AD EXISTS

Enable AD DS authentication on the storage account

CLOUD-ONLY / ENTRA ID

Enable Entra ID Kerberos authentication

SIMPLE / DEV

Storage account key (not for production)

DECIDE

Is this a hybrid setup with on-prem Windows Servers?

YES — EXTEND ON-PREM

Deploy Azure File Sync: install agent → register server → create Sync Group → add server endpoint

CLOUD-ONLY

Mount the share directly via SMB or NFS

DECIDE

Should the share be reachable from the public internet?

NO (RECOMMENDED)

Deploy a Private Endpoint + disable public access on the account

YES (TEMPORARY / DEV)

Open port 445 outbound; confirm ISP doesn't block it

DONE

Share deployed, authenticated, and connected — on-prem or cloud

Tools for configuring Azure Files

File shares, sync agents, snapshots, and mount commands — here's which tool to reach for and the exact commands you need.

Azure Portal

BEST FOR: share creation, snapshot management, File Sync setup

The portal has a "Connect" button that generates the exact mount script for Windows, Linux, or macOS — fastest way to get the right command for a share.

```
Storage Account → File shares
  → + File share → Connect (auto-generates mount script)
  → Azure File Sync (Storage Sync Services)
```

Azure CLI

BEST FOR: scripted share creation and snapshot management

Create shares, set quotas, and manage snapshots in commands that drop into deployment pipelines.

```
az storage share create \
  --account-name "myaccount" \
  --name "myshare" --quota 100

az storage share snapshot \
  --account-name "myaccount" --name "myshare"

az storage share list \
  --account-name "myaccount" --output table
```

>_ Mount — Windows

`BEST FOR: mapping a drive letter to the share`

Standard `net use` maps the share as a drive. Port 445 must be open outbound (often blocked on home/hotel ISPs).

```
net use Z:
\\myaccount.file.core.windows.net\m
yshare `
  /user:AZURE\myaccount
<storageAccountKey> /persistent:yes

# Or via PowerShell
$cred = Get-Credential
New-PSDrive -Name Z -PSProvider
FileSystem `
  -Root
"\\myaccount.file.core.windows.net\
myshare" `
  -Credential $cred -Persist
```

>_ Mount — Linux

`BEST FOR: mounting SMB shares on Linux VMs`

Uses the `cifs-utils` package. Add to `/etc/fstab` for persistent mounts across reboots.

```
sudo apt install cifs-utils -y
sudo mkdir /mnt/myshare
sudo mount -t cifs \

//myaccount.file.core.windows.net/m
yshare \
  /mnt/myshare \
  -o username=myaccount,password=
<key>,vers=3.0
```

‹› Bicep / ARM

`BEST FOR: IaC — file shares as part of deployment`

Deploy shares with quota and tier settings alongside the rest of the storage account in a version-controlled template.

```
resource fileShare
'Microsoft.Storage/storageAccounts/
fileServices/shares@2023-01-01' = {
  name: 'myshare'
  parent: fileService
  properties: {
    shareQuota: 100
    accessTier: 'Hot'
    enabledProtocols: 'SMB'
  }
}
```

TOOL	BEST FOR	SUPPORTS FILE SYNC?
Portal	Share creation, mount script generation, sync setup	**Yes**
CLI	Scripted share + snapshot management	Partial
PowerShell (Az)	Windows-centric automation	**Yes**
Bicep / ARM	IaC share deployment	No
File Sync Agent	On-prem server registration and sync	**Yes — required**

Principles worth internalizing

Use identity-based auth over storage keys, lock down port 445 via private endpoints, and always test File Sync cloud tiering recall behavior before going live.

AUTH METHOD	WORKS WITH SMB?	WORKS WITH NFS?	REQUIRES ON-PREM AD?
Storage Account Key	**Yes**	No	No
On-prem AD DS (Kerberos)	**Yes**	No	**Yes**
Entra ID Kerberos	**Yes**	No	No
Entra Domain Services	**Yes**	No	No (managed)

EXAM GOTCHA — PORT 445

SMB uses **port 445** outbound. Many ISPs (residential, hotel, campus networks) block port 445, which will silently fail a mount. The exam tests that you know this is a common cause of mount failures for Azure Files over SMB. The fix is either using a **VPN/ExpressRoute** or a **private endpoint**.

FILE SYNC VS. DIRECT MOUNT

Use **Azure File Sync** when the server needs a local cache of hot data for performance, or when users on-premises must access files even if Azure is temporarily unreachable. Use a **direct mount** when the workload runs in Azure and a shared drive is all that's needed — simpler, no agent to maintain.

Six mistakes that show up on the exam

Trying to create an NFS share on a GPv2 account

NFS requires a Premium FileStorage account — GPv2 supports SMB only.

Forgetting port 445 is often blocked

SMB mounts fail silently on networks where ISPs block port 445 outbound. Use VPN, ExpressRoute, or a private endpoint.

Confusing share-level and file-level permissions

Both RBAC (share-level) and NTFS ACLs (file-level) are evaluated — a user needs both to access a file.

Relying on snapshots as backups

Snapshots are incremental and stored in the same account — they don't protect against account-level deletion. Use Azure Backup for file shares.

Cloud tiering stubs failing on network loss

Stub files recall transparently from Azure — if network connectivity drops, opening a tiered file will fail. Size the local cache appropriately for working set.

Using storage key auth in production SMB

The account key grants full access to all shares — use identity-based auth (AD DS or Entra ID Kerberos) and RBAC share-level roles instead.